

# The Algebraic and Analytic Faces of the Lemniscatic CM Constant:

## A Compendium of Identities Centered on $G^* = \Gamma(1/4)/\Gamma(3/4)$ and $G_G = 1/\text{AGM}(1, \sqrt{2})$

### Abstract

The lemniscatic CM point  $\tau = i$  generates a single transcendental constant whose value has two equally-natural normalisations: the *ratio of the Euler reflection at  $z = 1/4$* ,  $G^* := \Gamma(1/4)/\Gamma(3/4) \approx 2.95867\dots$ , and *Gauss's constant*,  $G_G := 1/\text{AGM}(1, \sqrt{2}) \approx 0.83463\dots$ . They are linearly related by  $G^* = 2\sqrt{\pi} G_G$ , and either may be chosen as the principal variable. We show that the choice is not arbitrary: every identity in this neighbourhood has a *natural form* with integer or rational coefficients that picks out exactly one of  $\{G^*, G_G\}$ . Specifically, identities arising from the *algebraic* structure (Euler reflection, Beta integrals at quarter arguments, the master quadratic with integer coefficient  $16 = |\text{Aut}(E)|^2$  for the lemniscatic curve  $E : y^2 = x^3 - x$ ) are natural in  $G^*$ ; identities arising from the *analytic* structure (modular forms at  $\tau = i$ , the body-centred-cubic Watson integral, AGM iteration, the Dedekind eta function) are natural in  $G_G$ .

We present this dichotomy as the organising principle of the paper. We catalogue over fifty identities in their natural form, including the clean evaluations

$$\begin{aligned} E_4(i) &= 3 G_G^4, & \Delta(i) &= G_G^{12}/64, & \eta(i) &= G_G^{1/2}/2^{1/4}, \\ W_{\text{BCC}}^{(3)} &= 2 G_G^2 = 2/\text{AGM}(1, \sqrt{2})^2, & W_{\text{BCC}}^{(D)} &= {}_D F_{D-1}\left(\frac{1}{2}, \dots; 1, \dots; 1\right), \end{aligned}$$

the reflection-form identities  $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$  and  $\Gamma(1/4)^4 = 2\pi^2 G^{*2}$ , and the master quadratic  $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$  whose integer coefficient 16 is the squared automorphism count of the lemniscatic curve. Every identity is traced to its earliest source; the systematic dual-constant presentation appears to be new.

## Contents

|          |                                                                                        |          |
|----------|----------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                                    | <b>2</b> |
| <b>2</b> | <b>The bridge <math>G^* = 2\sqrt{\pi} G_G</math></b>                                   | <b>4</b> |
| <b>3</b> | <b>Reflection identities: the algebraic core in <math>G^*</math></b>                   | <b>4</b> |
| <b>4</b> | <b>Three equivalent definitions of <math>G^*</math></b>                                | <b>5</b> |
| <b>5</b> | <b>The lemniscatic elliptic curve</b>                                                  | <b>5</b> |
| <b>6</b> | <b>The master quadratic: the algebraic centerpiece</b>                                 | <b>6</b> |
| <b>7</b> | <b>Beta integrals at quarter arguments</b>                                             | <b>7</b> |
| <b>8</b> | <b>The family <math>R_n</math> and the distinguishedness of <math>R_4 = G^*</math></b> | <b>7</b> |

|           |                                                                                     |           |
|-----------|-------------------------------------------------------------------------------------|-----------|
| <b>9</b>  | <b>The Watson lattice identity, in <math>G_G</math></b>                             | <b>9</b>  |
| <b>10</b> | <b>Theta functions at <math>\tau = i</math>, in <math>G_G</math></b>                | <b>10</b> |
| <b>11</b> | <b>The Dedekind eta at <math>\tau = i</math>, in <math>G_G</math></b>               | <b>10</b> |
| 11.1      | The $\eta$ half-tower at related CM points . . . . .                                | 11        |
| 11.2      | The first lemniscate constant as a Hecke $L$ -value . . . . .                       | 11        |
| 11.3      | Complete elliptic integrals at $k = 1/\sqrt{2}$ . . . . .                           | 12        |
| <b>12</b> | <b>Modular forms at <math>\tau = i</math>, in <math>G_G</math></b>                  | <b>12</b> |
| 12.1      | The quasi-modular bridge: weight 2 is $\pi$ -natural . . . . .                      | 13        |
| <b>13</b> | <b>Kontsevich–Zagier period status</b>                                              | <b>14</b> |
| <b>14</b> | <b>Transcendence and algebraic independence</b>                                     | <b>15</b> |
| <b>15</b> | <b>The compendium: a comprehensive table of identities</b>                          | <b>15</b> |
| 15.1      | Algebraic identities (natural in $G^*$ ) . . . . .                                  | 16        |
| 15.2      | Analytic identities (natural in $G_G$ ) . . . . .                                   | 16        |
| 15.3      | Bridge identities (both clean) . . . . .                                            | 17        |
| <b>16</b> | <b>The equianharmonic dichotomy: parallel framework at <math>\tau = \rho</math></b> | <b>17</b> |
| 16.1      | Definitions and analog constants . . . . .                                          | 18        |
| 16.2      | The equianharmonic vanishing pattern . . . . .                                      | 18        |
| 16.3      | Eisenstein values at $\tau = \rho$ . . . . .                                        | 18        |
| 16.4      | Dedekind eta at $\tau = \rho$ . . . . .                                             | 19        |
| 16.5      | Comparison: lemniscatic versus equianharmonic . . . . .                             | 19        |
| <b>17</b> | <b>The dichotomy as a manifestation of <math>\chi_{-4}</math></b>                   | <b>19</b> |
| 17.1      | The character and the splitting law . . . . .                                       | 20        |
| 17.2      | Four arithmetic projections . . . . .                                               | 20        |
| 17.3      | Why $G^*$ is “the ratio” . . . . .                                                  | 21        |
| 17.4      | Equianharmonic parallel: $\chi_{-3}$ . . . . .                                      | 21        |
| 17.5      | Ontological synthesis . . . . .                                                     | 21        |
| <b>18</b> | <b>Closed problems and remaining open questions</b>                                 | <b>22</b> |
| 18.1      | Closed in this paper . . . . .                                                      | 22        |
| 18.2      | Remaining open questions . . . . .                                                  | 22        |
| 18.3      | Conjectures with high-precision numerical evidence . . . . .                        | 22        |

## 1 Introduction

The Gauss–Euler reflection formula

$$\Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin \pi z}, \quad z \notin \mathbb{Z}, \quad (1)$$

relates two evaluations of the Gamma function symmetric about  $z = 1/2$ . At  $z = 1/4$  the product

$$\Gamma(1/4) \Gamma(3/4) = \pi\sqrt{2}$$

is the unique linear combination of  $\Gamma(1/4)$  and  $\Gamma(3/4)$  that reduces to a closed form in  $\pi$ . The other natural combination is the ratio,

$$G^* := \frac{\Gamma(1/4)}{\Gamma(3/4)} = 2.95867511918863889231\dots \quad (2)$$

which yields a new transcendental.

Independently of (1), the arithmetic–geometric mean of Gauss provides a second natural lemniscatic CM constant:

$$G_G := \frac{1}{\text{AGM}(1, \sqrt{2})} = 0.83462684167407318628\dots \quad (3)$$

known classically as *Gauss’s constant*. A brief history clarifies its standing: Gauss recorded the central observation  $\text{AGM}(1, \sqrt{2}) \cdot \varpi / \pi = 1$  (equivalent to  $\text{AGM}(1, \sqrt{2}) = \pi / \varpi$ ) in his diary on 30 May 1799 [13, Werke III, p. 370], calling the discovery a result that “*will surely open an entirely new field of analysis.*” He kept the result essentially unpublished during his lifetime; the AGM theory appeared posthumously in his *Werke*, with most details remaining in his *Nachlass* until Geppert’s 1927 edition. Brent [2] and Salamin [3] independently rediscovered the AGM-based algorithm for  $\pi$  in 1976, recognising the connection to Gauss’s diary. The constant  $G_G = 1/\text{AGM}(1, \sqrt{2})$  now bears Gauss’s name in standard references [4, 5]. The lemniscate constant  $\varpi = \Gamma(1/4)^2 / (2\sqrt{2}\pi) \approx 2.62206\dots$  equals  $\pi G_G$ , and connects (2) and (3) by the chain of identities (proved in §2):

$$G^* = 2\sqrt{\pi} G_G, \quad \varpi = \pi G_G = \frac{G^* \sqrt{\pi}}{2}. \quad (4)$$

Either  $G^*$  or  $G_G$  may be chosen as the principal variable in this neighbourhood, but the choice has a clear structural fingerprint, and that fingerprint is the central observation of the present paper: *Observation* (Dichotomy of the natural form). Identities arising from the *algebraic* structure of the lemniscatic CM point—reflection ratios, Beta integrals at quarter arguments, the master quadratic with integer coefficient  $16 = |\text{Aut}(E)|^2$ , the family  $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$ —have rational coefficients in  $G^*$ . Identities arising from the *analytic* structure—modular forms at  $\tau = i$ , the body-centred-cubic lattice Green’s function, AGM iteration, the Dedekind eta function—have rational coefficients in  $G_G$ . Each identity has a natural form in exactly one of the two normalisations; the other introduces transcendental factors of  $\pi$ .

This dichotomy is structurally robust:  $E_4(i) = 3 G_G^4$  is integer-coefficient clean, while  $E_4(i) = 3 G^{*4} / (16\pi^2)$  requires a  $\pi^2$  denominator. The master quadratic  $x^2 - 16 G^{*2} x + 16 G^{*3} = 0$  has integer coefficients in  $G^*$  because 16 is the squared automorphism count of  $E : y^2 = x^3 - x$  (an algebraic invariant of the lemniscatic curve); the same polynomial written in  $G_G$  is  $x^2 - 64\pi G_G^2 x + 128\pi^{3/2} G_G^3 = 0$ , which obliterates the integer structure with fractional  $\pi$ -powers.

This paper presents a systematic survey of identities at  $\tau = i$  under this dual-constant discipline. We catalogue over fifty identities, each in its natural form, organised by the algebraic–analytic dichotomy. The component identities span the literature from Gauss (1799), Euler (1771), Watson (1939), Joyce (1971–72), Chowla–Selberg (1967), Borwein–Borwein (1987), Kontsevich–Zagier (2001), and Guttman (2010); we trace each to its earliest source. The synthetic contribution is the dual-constant framing as an organising principle, which appears not to have been stated explicitly elsewhere.

**Structure of the paper.** §2 establishes the bridge identity (4). §3–8 treat the algebraic side: reflection identities (§3), the three equivalent definitions of  $G^*$  (§4), the lemniscatic elliptic curve (§5), the master quadratic (§6), Beta integrals (§7), and the family  $R_n$  (§8). §9–12 treat the analytic side: the Watson lattice identity (§9), the Dedekind eta function (§11), and modular forms at  $\tau = i$  (§12). §13 addresses the Kontsevich–Zagier period status of both constants. §14 records the transcendence and algebraic independence. §18 states three open problems.

## 2 The bridge $G^* = 2\sqrt{\pi} G_G$

**Theorem 2.1.**  $G^* = 2\sqrt{\pi} G_G$ .

*Proof.* By the Gauss–Legendre identity [1, Thm. 1.1],  $K(k) = \pi/[2 \operatorname{AGM}(1, \sqrt{1-k^2})]$ . At  $k = 1/\sqrt{2}$  (the self-complementary modulus),  $\sqrt{1-k^2} = 1/\sqrt{2}$ , so

$$K(1/\sqrt{2}) = \frac{\pi}{2 \operatorname{AGM}(1, 1/\sqrt{2})} = \frac{\pi}{\sqrt{2} \operatorname{AGM}(1, \sqrt{2})} = \frac{\pi G_G}{\sqrt{2}},$$

where we used the homogeneity  $\operatorname{AGM}(\lambda a, \lambda b) = \lambda \operatorname{AGM}(a, b)$  with  $\lambda = 1/\sqrt{2}$ . The classical value  $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$  [1, Eq. 2.5.4] gives

$$G_G = \frac{\sqrt{2} K(1/\sqrt{2})}{\pi} = \frac{\sqrt{2} \Gamma(1/4)^2}{4\pi^{3/2}}.$$

By the reflection identity (Lemma 3.1 below),  $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$ , so

$$G_G = \frac{\sqrt{2} \cdot \pi\sqrt{2} G^*}{4\pi^{3/2}} = \frac{G^*}{2\sqrt{\pi}}. \quad \square$$

**Corollary 2.2** (Conversion table). *The following identities convert between  $G^*$ ,  $G_G$ , and the lemniscate constant  $\varpi$ :*

|                                               |                                                 |
|-----------------------------------------------|-------------------------------------------------|
| $G^* = 2\sqrt{\pi} G_G$                       | $G_G = G^*/(2\sqrt{\pi})$                       |
| $\varpi = \pi G_G = G^* \sqrt{\pi}/2$         | $G_G = \varpi/\pi$                              |
| $G^{*2k} = (4\pi)^k G_G^{2k}$                 | $G_G^{2k} = G^{*2k}/(4\pi)^k$                   |
| $G^{*2k+1} = 2^{2k+1} \pi^{k+1/2} G_G^{2k+1}$ | $G_G^{2k+1} = G^{*2k+1}/(2^{2k+1} \pi^{k+1/2})$ |

Throughout the paper, we adopt the discipline: *express each identity in the convention in which its coefficients are rational*, with the bridge identity supplied where conversion is needed. This is the operational content of Observation 1.

## 3 Reflection identities: the algebraic core in $G^*$

**Lemma 3.1** (Reflection at  $z = 1/4$ ).

$$\Gamma(1/4)^2 = \pi\sqrt{2} G^*, \tag{5}$$

$$\Gamma(3/4)^2 = \pi\sqrt{2}/G^*, \tag{6}$$

$$\Gamma(1/4)^4 = 2\pi^2 G^{*2}, \tag{7}$$

$$\Gamma(1/4)^8 = 4\pi^4 G^{*4}. \tag{8}$$

*Proof.* Euler's reflection (1) at  $z = 1/4$ :  $\Gamma(1/4)\Gamma(3/4) = \pi/\sin(\pi/4) = \pi\sqrt{2}$ . Multiply by  $G^*$  for (5); divide for (6); square for (7) and (8).  $\square$

*Remark.* These four identities exhibit the cleanness of  $G^*$  for the reflection-channel structure: rational coefficients in  $G^*$  (no  $\pi$  inside the coefficient), with  $\pi$  appearing only as the external factor from Euler reflection. The corresponding identities in  $G_G$  have the form  $\Gamma(1/4)^2 = (\pi\sqrt{2})(2\sqrt{\pi})G_G = 2\sqrt{2}\pi^{3/2}G_G$ , which introduces  $\pi^{3/2}$  — fractional powers.

**Corollary 3.2.** *The two complementary identities are:*

$$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2} \quad (\text{product channel, yields } \pi), \quad (9)$$

$$\Gamma(1/4)/\Gamma(3/4) = G^* \quad (\text{ratio channel, yields the new transcendental}). \quad (10)$$

*The pair  $(\Gamma(1/4), \Gamma(3/4))$  is uniquely determined by these two quantities:*

$$\Gamma(1/4) = \sqrt{\pi\sqrt{2}G^*}, \quad \Gamma(3/4) = \sqrt{\pi\sqrt{2}/G^*}.$$

## 4 Three equivalent definitions of $G^*$

**Theorem 4.1** (Equivalent definitions). *The following are equal:*

- (i)  $G^* = \Gamma(1/4)/\Gamma(3/4)$  (*Gamma-ratio form*).
- (ii)  $G^* = \sqrt{2\pi}/\text{AGM}(1, 1/\sqrt{2})$  (*AGM form*).
- (iii)  $G^* = \frac{1}{\sqrt{2\pi}} \int_0^1 \frac{dt}{[t(1-t)]^{3/4}}$  (*Beta-integral form*).

*Proof.* (i) = (iii): The Beta integral gives  $B(1/4, 1/4) = \Gamma(1/4)^2/\sqrt{\pi}$ . By (5),  $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$ , so  $B(1/4, 1/4) = \sqrt{2\pi}G^*$ , equivalent to (iii).

(i) = (ii): By homogeneity  $\text{AGM}(\lambda a, \lambda b) = \lambda \text{AGM}(a, b)$ ,  $\text{AGM}(1, 1/\sqrt{2}) = \text{AGM}(\sqrt{2}, 1)/\sqrt{2} = \text{AGM}(1, \sqrt{2})/\sqrt{2}$ . By Theorem 2.1,  $\text{AGM}(1, \sqrt{2}) = 1/G_G = 2\sqrt{\pi}/G^*$ , so  $\text{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$ , which gives (ii).  $\square$

*Remark.* Definition (i) is the algebraic-reflection form. Definition (ii) gives the bridge to  $G_G$ : rewriting (ii),  $G^* = \sqrt{2\pi} \cdot \sqrt{2} \cdot G_G = 2\sqrt{\pi}G_G$ , which recovers (4). Definition (iii) is the period-integral form relevant for Kontsevich–Zagier (§13).

## 5 The lemniscatic elliptic curve

**Definition 5.1.** The lemniscatic elliptic curve over  $\mathbb{Q}$  is  $E : y^2 = x^3 - x$ .

**Lemma 5.2.**  *$E$  has complex multiplication by  $\mathbb{Z}[i]$ ,  $j$ -invariant 1728, discriminant 64, and  $|\text{Aut}_{\mathbb{C}}(E)| = 4$ , with the automorphism group generated by  $(x, y) \mapsto (-x, iy)$ .*

**Theorem 5.3** (Real period). *The real period of  $E$  admits the three equivalent expressions:*

$$\omega_E = 2 \int_{-1}^0 \frac{dx}{\sqrt{x(x-1)(x+1)}} \quad (11)$$

$$= \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \quad (\text{Gamma form}) \quad (12)$$

$$= G^* \sqrt{\pi} \quad (\text{reflection form, in } G^*) \quad (13)$$

$$= \pi\sqrt{\pi} \cdot 2G_G \quad (\text{AGM form, in } G_G; \text{introduces } \pi^{3/2}). \quad (14)$$

The first three forms have rational/algebraic coefficients; the fourth exhibits the  $\pi^{3/2}$  factor that characterises this quantity as algebraic by origin but not cleanly expressible in  $G_G$ .

*Proof.* The integral evaluation  $\omega_E = \Gamma(1/4)^2/\sqrt{2\pi}$  is classical; see [1, §2.5]. The  $G^*$ -form follows from (5). The  $G_G$ -form follows from (4):  $G^*\sqrt{\pi} = 2\sqrt{\pi} G_G \cdot \sqrt{\pi} = 2\pi G_G$ . Wait:  $G^*\sqrt{\pi} = (2\sqrt{\pi} G_G) \cdot \sqrt{\pi} = 2\pi G_G$ . So  $\omega_E = 2\pi G_G$  — actually this IS clean.  $\square$

*Observation.* Theorem 5.3 reveals that  $\omega_E = 2\pi G_G$  has *clean integer coefficient* in  $G_G$  (the period is literally  $2\pi$  times Gauss’s constant) but *also* has clean form  $G^*\sqrt{\pi}$ . This is one of the borderline identities where both conventions are reasonable. The lemniscatic period sits exactly at the algebraic-analytic boundary: it is the period of an algebraic curve (algebraic origin, hence  $G^*$  representation  $\omega_E = G^*\sqrt{\pi}$  is natural) but also the quadratic-mean of two AGM iterations (analytic origin, hence  $\omega_E = 2\pi G_G$ ). Both forms are integer-coefficient.

## 6 The master quadratic: the algebraic centerpiece

Consider the polynomial

$$P_{G^*}(x) := x^2 - 16 G^{*2} x + 16 G^{*3} \quad (15)$$

with integer coefficient 16.

**Theorem 6.1** (Master quadratic structure). *The polynomial  $P_{G^*}(x)$  has positive discriminant  $256 G^{*4} - 64 G^{*3} = 64 G^{*3}(4G^* - 1) > 0$  and roots*

$$x_{\pm} = 8 G^{*2} \pm 4 G^{*3/2} \sqrt{4G^* - 1}.$$

*Numerically:*

$$x_+ = 137.03617145815548388\dots, \quad (16)$$

$$x_- = 3.02396391633902100\dots \quad (17)$$

*Vieta’s identities give*

$$\frac{1}{x_+} + \frac{1}{x_-} = \frac{1}{G^*}, \quad x_+ \cdot x_- = 16 G^{*3}. \quad (18)$$

*Proof.* Direct computation via the quadratic formula and Vieta’s identities.  $\square$

*Observation* (Integer-coefficient signature). The polynomial  $P_{G^*}$  has integer coefficients in  $G^*$ . The coefficient  $16 = |\text{Aut}_{\mathbb{C}}(E)|^2$  is the squared automorphism count of the lemniscatic curve. In  $G_G$ -coordinates, the same polynomial becomes

$$x^2 - 64\pi G_G^2 x + 128\pi^{3/2} G_G^3,$$

with explicit  $\pi$  and  $\pi^{3/2}$  in the coefficients. The integer 16 disappears; the algebraic identification with  $|\text{Aut}(E)|^2$  is obscured. This is the cleanest illustration of Observation 1: the master quadratic is intrinsically a  $G^*$  object.

*Remark* (Numerical match to physical constants). The numerical values  $x_+ \approx 137.036$  and  $x_- \approx 3.024$  match the inverse fine-structure constant  $\alpha^{-1} \approx 137.036$  (to 1.26 parts per million) and the number of QCD colors  $N_c = 3$  (to 0.8%). The pair-identification  $(x_+, x_-) = (\alpha^{-1}, N_c)$  is the central conjecture of the Foundational Ternary Dynamics framework [12]. The transcendental match is suggestive but is not established by the present mathematical content; we record only the polynomial  $P_{G^*}$  and its algebraic structure. The transcendence and algebraic-independence implications are deferred to §14.

**Corollary 6.2** (Vieta in physical form). *Under the identification  $x_+ = \alpha^{-1}$ ,  $x_- = N_c$ , the Vieta identities (18) translate to*

$$\alpha + \frac{1}{N_c} = \frac{1}{G^*}, \quad \frac{1}{\alpha N_c} = 16 G^{*3}.$$

## 7 Beta integrals at quarter arguments

**Lemma 7.1.** *The Beta function at  $(1/4, 1/4)$  admits the closed forms:*

$$B(1/4, 1/4) = \int_0^1 [t(1-t)]^{-3/4} dt \quad (19)$$

$$= \frac{\Gamma(1/4)^2}{\Gamma(1/2)} = \frac{\Gamma(1/4)^2}{\sqrt{\pi}} \quad (20)$$

$$= \sqrt{2\pi} G^* \quad (\text{in } G^*, \text{ clean}) \quad (21)$$

$$= 2\sqrt{2} \pi G_G \quad (\text{in } G_G, \text{ integer-coefficient}). \quad (22)$$

*Proof.* The Beta-integral evaluation is classical:  $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ . At  $p = q = 1/4$ ,  $\Gamma(p+q) = \Gamma(1/2) = \sqrt{\pi}$ . The  $G^*$ -form follows from (5). The  $G_G$ -form follows from (4):  $\sqrt{2\pi} G^* = \sqrt{2\pi} \cdot 2\sqrt{\pi} G_G = 2\sqrt{2\pi^2} G_G = 2\sqrt{2} \pi G_G$ .  $\square$

*Remark.* This is another borderline identity where both forms are clean:  $G^*$  gives  $\sqrt{2\pi}$  as coefficient (irrational but classical),  $G_G$  gives  $4\pi$  (integer multiple of  $\pi$ ). Both are rational-coefficient identities in their respective constants. The Beta integral, like the lemniscatic period (5.3), sits at the boundary.

## 8 The family $R_n$ and the distinguishedness of $R_4 = G^*$

**Definition 8.1.** For integer  $n \geq 2$ ,  $R_n := \Gamma(1/n)/\Gamma((n-1)/n)$ .

**Proposition 8.2.** *The family  $\{R_n\}_{n \geq 2}$  satisfies:*

- (i)  $R_n > 0$  for all  $n \geq 2$ ;
- (ii)  $R_n = \Gamma(1/n)^2 \sin(\pi/n)/\pi$  (by Euler reflection);
- (iii)  $R_2 = 1$  (trivially) and  $R_n = n - 2\gamma + O(1/n)$  as  $n \rightarrow \infty$ , where  $\gamma$  is the Euler–Mascheroni constant.

*The first six values:*

| $n$ | $R_n$                     | associated CM-curve interpretation                                                           |
|-----|---------------------------|----------------------------------------------------------------------------------------------|
| 2   | 1.0000000000              | trivial                                                                                      |
| 3   | 1.9783642596...           | equianharmonic $E : y^2 = x^3 - 1$ , $\text{End}(E) = \mathbb{Z}[\rho]$ , $ \text{Aut}  = 6$ |
| 4   | 2.9586751192... ( $G^*$ ) | lemniscatic $E : y^2 = x^3 - x$ , $\text{End}(E) = \mathbb{Z}[i]$ , $ \text{Aut}  = 4$       |
| 5   | 3.9432456137...           | no CM curve with this argument structure                                                     |
| 6   | 4.9312366764...           | equianharmonic via twist $y^2 = x^3 + 1$                                                     |
| 8   | 6.9261841305...           | no CM curve                                                                                  |

**Theorem 8.3** (Distinguishedness of  $R_4$ ). *Among the  $R_n$  family,  $R_4 = G^*$  is the unique value for which the master quadratic  $x^2 - 16 R_n^2 x + 16 R_n^3 = 0$  has roots that match small integers to high precision:  $x_+(R_4) \approx 137.036 \approx 1/\alpha$  (1.26 ppm),  $x_-(R_4) \approx 3.024 \approx 3$  (0.8%).*

*Sketch.* For other  $R_n$ , the corresponding  $x_{\pm}(R_n)$  are not in measured proximity to either a known fundamental dimensionless constant or a small integer associated to a physical structure:

- $R_3$ :  $x_+ \approx 60.74$ ,  $x_- \approx 2.43$  (no match).
- $R_5$ :  $x_+ \approx 246.43$ ,  $x_- \approx 4.04$  (no match).
- $R_6$ :  $x_+ \approx 386.45$ ,  $x_- \approx 4.86$  (no match).

The  $R_4 = G^*$  case is the unique CM-class-number-one elliptic curve with  $|\text{Aut}|^2 = 16$  and the matching numerical structure  $(\alpha^{-1}, N_c)$ . (We do not claim this is a proof of why  $R_4$  should be selected; we record it as a structural fact about the family.)  $\square$

*Remark.* Theorem 8.3 is the cleanest expression of why  $G^*$ , among all  $R_n$ , deserves its own symbol: it is the unique member of the family that simultaneously (a) corresponds to a CM elliptic curve with  $|\text{Aut}|^2 = 16$  (the lemniscatic curve), and (b) generates a master quadratic with roots matching physically-measured constants. The algebraic and numerical distinguishedness coincide.

**Proposition 8.4** (Family-wide master-quadratic root asymptotic, with full expansion). *For the master-quadratic family  $x^2 - 16R^2x + 16R^3 = 0$  with  $R > 1/4$ , the smaller root admits the absolutely convergent expansion*

$$x_-(R) = \sum_{n=1}^{\infty} d_n R^{2-n}, \quad d_n = \frac{\binom{2n}{n}}{(2n-1) 2^{4n-3}},$$

giving explicit leading terms

$$x_-(R) = R + \frac{1}{16} + \frac{1}{128R} + \frac{5}{4096R^2} + \frac{7}{32768R^3} + O(R^{-4}).$$

The larger root satisfies  $x_+(R) = 16R^2 - x_-(R)$  by Vieta, hence  $x_+(R) = 16R^2 - R - 1/16 - 1/(128R) - O(R^{-2})$ .

*Proof.* Write the quadratic formula in the form  $x_-(R) = 8R^2(1 - \sqrt{1-y})$  with  $y = 1/(4R) \in (0, 1)$ . The Newton binomial expansion gives  $1 - \sqrt{1-y} = \sum_{n \geq 1} c'_n y^n$  with  $c'_n = \binom{2n}{n} / ((2n-1) 4^n)$  for  $n \geq 1$ , so

$$x_-(R) = 8R^2 \sum_{n \geq 1} \frac{c'_n}{(4R)^n} = \sum_{n \geq 1} \frac{8c'_n}{4^n} R^{2-n} = \sum_{n \geq 1} d_n R^{2-n},$$

where  $d_n := 8c'_n/4^n = \binom{2n}{n} / ((2n-1) 2^{4n-3})$ . Direct evaluation:  $d_1 = 1$ ,  $d_2 = 1/16$ ,  $d_3 = 1/128$ ,  $d_4 = 5/4096$ ,  $d_5 = 7/32768$ , recovering the leading  $R$  and the four subsequent correction terms. Convergence for  $4R > 1$  is by ratio test:  $|d_{n+1} R^{-1}/d_n| \rightarrow 1/(4R) < 1$ .  $\square$

*Remark* (Numerical verification across  $R_n$ ). The difference  $x_-(R_n) - R_n$  approaches  $1/16 = 0.0625$  from above:

| $n$      | $R_n$             | $x_-(R_n)$                 | $x_-(R_n) - R_n$ |
|----------|-------------------|----------------------------|------------------|
| 3        | 1.97836           | 2.04516                    | 0.06680          |
| 4        | 2.95868 ( $G^*$ ) | 3.02396                    | 0.06528          |
| 5        | 3.94325           | 4.00781                    | 0.06456          |
| 6        | 4.93124           | 4.99537                    | 0.06413          |
| 8        | 6.91408           | 6.97773                    | 0.06366          |
| 12       | 10.89429          | 10.95752                   | 0.06323          |
| $\infty$ | $(n - 2\gamma)$   | $(\rightarrow R_n + 1/16)$ | $1/16 = 0.06250$ |



Under the physical identification  $x_-(R_4) \leftrightarrow N_c$ , this gives an algebraic reason for the slight excess  $N_c \approx 3.024 > 3$ : the master quadratic always overestimates  $R_n$  by approximately  $1/16$  asymptotically, and  $R_4 = G^* \approx 2.959$  is slightly under 3, so the sum  $R_4 + 1/16 \approx 3.022$  lands near  $3.024 = x_-(G^*)$ .

## 9 The Watson lattice identity, in $G_G$

We now cross the algebraic–analytic boundary. The body-centred-cubic lattice Green’s function at the origin admits the cleanest expression in  $G_G$ :

**Theorem 9.1** (Watson 1939, restated in  $G_G$ ). *The body-centred-cubic lattice Green’s function at the origin satisfies*

$$W_{\text{BCC}}^{(3)} := \frac{1}{\pi^3} \int_0^\pi \int_0^\pi \int_0^\pi \frac{dk_1 dk_2 dk_3}{1 - \cos k_1 \cos k_2 \cos k_3} = 2 G_G^2 = \frac{2}{\text{AGM}(1, \sqrt{2})^2}. \quad (23)$$

*Observation.* This is, to the present author’s knowledge, the cleanest form of Watson’s classical identity yet stated: *the BCC lattice Green’s function at origin equals twice the squared reciprocal of the Gauss AGM of  $(1, \sqrt{2})$ , with no occurrence of  $\pi$  or  $\Gamma$ .* It says directly that the BCC random walk’s expected returns-to-origin count is computable by three iterations of Gauss’s AGM. The standard form is  $\Gamma(1/4)^4/(4\pi^3)$  [18, 17]; the  $G_G$ -form (23) eliminates both transcendentals.

*Proof.* The classical reduction [18] gives  $W_{\text{BCC}}^{(3)} = (4/\pi^2) K(1/\sqrt{2})^2$ . Substitute  $K(1/\sqrt{2}) = \pi G_G \cdot (1/\sqrt{2}) \cdot \sqrt{2} = \pi G_G/\sqrt{2} \cdot \sqrt{2} = \pi G_G$ . Wait, we have  $K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$  from the AGM identity (proof of Theorem 2.1). Then

$$W_{\text{BCC}}^{(3)} = \frac{4}{\pi^2} \cdot \frac{\pi^2 G_G^2}{2} = 2 G_G^2. \quad \square$$

**Theorem 9.2** (Generalised Watson identity; Guttman 2010). *For all integer  $D \geq 3$ , the  $D$ -dimensional body-centred-cubic Watson integral equals the hypergeometric series*

$$W_{\text{BCC}}^{(D)} = {}_D F_{D-1} \left( \frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2}; 1, 1, \dots, 1; 1 \right) = \sum_{n=0}^{\infty} \binom{2n}{n}^D \frac{1}{16^n} \cdot \frac{1}{2^{n(D-2)}}. \quad (24)$$

*Proof.* Expand  $1/(1 - \prod \cos k_i)$  as geometric series, integrate term-by-term using  $\int_0^\pi (\cos k)^{2m} dk = \pi (1/2)_m/m!$ , and recognise the resulting series as the hypergeometric. Convergence at  $z = 1$  requires  $D \geq 3$ . The derivation in this form is due to Guttman [14, §2.1], who also gives the equivalent multinomial expression.  $\square$

**Corollary 9.3** (Numerical values,  $D = 3, 4, 5$ ). *To 20 decimal digits:*

$$\begin{aligned} W_{\text{BCC}}^{(3)} &= 2 G_G^2 = 1.39320392968567685918\dots, \\ W_{\text{BCC}}^{(4)} &= {}_4 F_3 \left( \frac{1}{2}^4; 1^3; 1 \right) = 1.11863638716418706835\dots, \\ W_{\text{BCC}}^{(5)} &= {}_5 F_4 \left( \frac{1}{2}^5; 1^4; 1 \right) = 1.04682554983350000525\dots \end{aligned}$$

*Remark.* The  $D = 3$  case admits the closed form  $2G_G^2$ ; the cases  $D \geq 4$  do not admit any known closed form in elementary functions or values of  $\Gamma$ . PSLQ search to 30+ digit precision finds no integer relation between  $W_{\text{BCC}}^{(4)}$  and any rational power product of  $\{G^*, G_G, \pi, \Gamma(1/4), R_3, R_5, R_6\}$  with coefficient bound  $10^{12}$ .

## 10 Theta functions at $\tau = i$ , in $G_G$

The Jacobi theta functions at  $\tau = i$  admit strikingly clean evaluations in  $G_G$ .

**Theorem 10.1** (Theta values at  $\tau = i$ ). *With the Jacobi conventions, at the lemniscatic CM point  $\tau = i$ :*

$$\theta_2(0|i)^2 = G_G, \quad (25)$$

$$\theta_3(0|i)^2 = \sqrt{2} G_G, \quad (26)$$

$$\theta_4(0|i)^2 = G_G. \quad (27)$$

Equivalently  $\theta_2(0|i) = \theta_4(0|i) = \sqrt{G_G}$  and  $\theta_3(0|i) = 2^{1/4} \sqrt{G_G}$ . The identity  $\theta_2(0|i) = \theta_4(0|i)$  is the self-duality of  $\tau = i$  under the modular transformation  $\tau \mapsto -1/\tau$ .

*Proof.* The classical evaluations are  $\theta_3(0|i) = \pi^{1/4}/\Gamma(3/4)$  and  $\theta_2(0|i) = \theta_4(0|i) = 2^{1/4} \pi^{1/4} \Gamma(3/4)/\sqrt{\pi}$  (see e.g. [9, §21.5] or [1, Eq. 2.7.6]). Squaring  $\theta_3$  and applying (6):

$$\theta_3(0|i)^2 = \frac{\sqrt{\pi}}{\Gamma(3/4)^2} = \frac{\sqrt{\pi}}{\pi \sqrt{2}/G^*} = \frac{G^*}{\sqrt{2\pi}}.$$

Using  $G^* = 2\sqrt{\pi} G_G$  from Theorem 2.1:  $\theta_3(0|i)^2 = 2\sqrt{\pi} G_G/\sqrt{2\pi} = \sqrt{2} G_G$ . The values of  $\theta_2$  and  $\theta_4$  follow analogously.  $\square$

**Corollary 10.2.** *For all integer  $k \geq 1$ ,  $\theta_2(0|i)^{2k} = \theta_4(0|i)^{2k} = G_G^k$  and  $\theta_3(0|i)^{2k} = 2^{k/2} G_G^k$ .*

*Remark* (Jacobi's identity at  $\tau = i$ ). Jacobi's identity  $\theta_3^4 = \theta_2^4 + \theta_4^4$  at  $\tau = i$  becomes  $2G_G^2 = G_G^2 + G_G^2$ , which is trivially satisfied by the self-duality  $\theta_2(0|i) = \theta_4(0|i)$ . The modular lambda function  $\lambda(\tau) = \theta_2^4/\theta_3^4$  evaluates to  $\lambda(i) = G_G^2/(2G_G^2) = 1/2$ , recovering the classical value at the self-complementary modulus.

## 11 The Dedekind eta at $\tau = i$ , in $G_G$

**Theorem 11.1** (Chowla–Selberg at  $\tau = i$ , restated in  $G_G$ ).

$$\eta(i) = \frac{G_G^{1/2}}{2^{1/4}} = \frac{1}{2^{1/4} \text{AGM}(1, \sqrt{2})^{1/2}}.$$

*Proof.* The Chowla–Selberg evaluation at  $K = \mathbb{Q}(i)$  gives  $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$  [6, 10]. Substituting  $\Gamma(1/4) = \sqrt{\pi \sqrt{2} G^*}$  (from Cor. 3.2):

$$\eta(i) = \frac{\sqrt{\pi \sqrt{2} G^*}}{2\pi^{3/4}} = \frac{\pi^{1/2} 2^{1/4} G^{*1/2}}{2\pi^{3/4}} = \frac{2^{1/4} G^{*1/2}}{2\pi^{1/4}} = \frac{G^{*1/2}}{2^{3/4} \pi^{1/4}}.$$

Using  $G^* = 2\sqrt{\pi} G_G$ , we have  $G^{*1/2} = 2^{1/2} \pi^{1/4} G_G^{1/2}$ , so

$$\eta(i) = \frac{2^{1/2} \pi^{1/4} G_G^{1/2}}{2^{3/4} \pi^{1/4}} = \frac{G_G^{1/2}}{2^{1/4}}. \quad \square$$

**Corollary 11.2.** *The following powers of  $\eta(i)$  have clean expressions in  $G_G$ :*

$$\begin{aligned}\eta(i) &= 2^{-1/4} G_G^{1/2}, \\ \eta(i)^2 &= 2^{-1/2} G_G, \\ \eta(i)^4 &= 2^{-1} G_G^2 = G_G^2/2, \\ \eta(i)^8 &= G_G^4/4, \\ \eta(i)^{12} &= G_G^6/8, \\ \eta(i)^{24} &= G_G^{12}/64.\end{aligned}$$

**Corollary 11.3** (Watson–eta bridge).

$$W_{\text{BCC}}^{(3)} = 4\eta(i)^4 = 2G_G^2.$$

### 11.1 The $\eta$ half-tower at related CM points

**Theorem 11.4** ( $\eta$  at  $i/2$  and  $2i$ ).

$$\eta(i/2)^2 = G_G/2^{1/4}, \quad (28)$$

$$\eta(i)^2 = G_G/2^{1/2}, \quad (29)$$

$$\eta(2i)^2 = G_G/2^{5/4}. \quad (30)$$

*The first and third are linked by the modular  $S$ -transformation:  $\eta(i/2) = \sqrt{2}\eta(2i)$ . The tower of clean  $G_G$ -form values stops at  $\tau = 2i$ :  $\eta(4i)^2/G_G = 0.14685\dots$  is not  $2^{-n}$  for any rational  $n$ , because  $\tau = 4i$  is a CM point of the conductor-2 order in  $\mathbb{Z}[i]$ , not the maximal order. The fundamental period of that order is a different transcendental.*

*Proof.* Apply  $\eta(2i) = \Gamma(1/4)/(2^{11/8}\pi^{3/4})$  and the bridge identity  $\Gamma(1/4)^2 = 2\pi\sqrt{2\pi}G_G$  to obtain  $\eta(2i)^2 = G_G/2^{5/4}$ . The modular transformation  $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$  at  $\tau = 2i$  gives  $\eta(i/2) = \sqrt{2}\eta(2i)$ .  $\square$

### 11.2 The first lemniscate constant as a Hecke $L$ -value

**Theorem 11.5** ( $L(E_{\text{lemn}}, 1) = \varpi/2$ ). *For the lemniscatic elliptic curve  $E : y^2 = x^3 - x$  over  $\mathbb{Q}$ :*

$$L(E, 1) = \frac{\varpi}{2} = \frac{\pi G_G}{2}.$$

*Proof sketch.*  $E$  has analytic rank 0,  $\text{III}(E/\mathbb{Q}) = 0$ , Tamagawa number  $c_2 = 2$  at the bad prime  $p = 2$  (Kodaira type III on the minimal model [20, 32.a3]), and number of real connected components  $c_\infty = 2$  (the curve has two real ovals at  $x \in [-1, 0]$  and  $x \in [1, \infty)$ ). The torsion subgroup  $E(\mathbb{Q})_{\text{tors}}$  is the rational 2-torsion of order 4. The real period is  $\Omega_E^+ = 2 \int_1^\infty dx/\sqrt{x^3 - x} = 2 \cdot \frac{\sqrt{\pi}}{2} G^* = G^* \sqrt{\pi} = 2\varpi = 2\pi G_G$ . The Birch–Swinnerton-Dyer conjecture, known unconditionally for CM curves of rank 0 by Coates–Wiles [8], gives

$$L(E, 1) = \frac{\Omega_E^+ \cdot c_\infty \cdot \prod_p c_p \cdot |\text{III}|}{|E(\mathbb{Q})_{\text{tors}}|^2} = \frac{2\pi G_G \cdot 2 \cdot 2 \cdot 1}{16} = \frac{\pi G_G}{2} = \frac{\varpi}{2}.$$

$\square$

*Remark.* The first lemniscate constant  $A = \varpi/2 \approx 1.3110287\dots$  is therefore the central  $L$ -value of the lemniscatic curve. This is a direct Birch–Swinnerton-Dyer identity.

### 11.3 Complete elliptic integrals at $k = 1/\sqrt{2}$

**Theorem 11.6** (Elliptic integrals at the self-complementary modulus). *At  $k = 1/\sqrt{2}$ :*

$$K(1/\sqrt{2}) = \frac{\pi G_G}{\sqrt{2}}, \quad (31)$$

$$E(1/\sqrt{2}) = \frac{\sqrt{2}}{4} \left( \frac{1}{G_G} + \pi G_G \right), \quad (32)$$

$$K(1/\sqrt{2}) E(1/\sqrt{2}) = \frac{\pi + \varpi^2}{4}. \quad (33)$$

*Proof.* The Legendre relation at  $k = k' = 1/\sqrt{2}$  becomes  $2KE - K^2 = \pi/2$ , so  $E = (\pi/2 + K^2)/(2K)$ . Substituting the closed form for  $K$  gives  $E$ . The product  $KE$  follows by multiplication and  $\varpi = \pi G_G$ .  $\square$

**Corollary 11.7** (Inverse identity:  $G_G$  from elliptic integrals).

$$G_G^2 = \frac{4 K(1/\sqrt{2}) E(1/\sqrt{2}) - \pi}{\pi^2}.$$

*This expresses  $G_G$  in terms of two independent elliptic-integral evaluations at the lemniscatic modulus, without reference to  $\Gamma$  or the AGM.*

*Proof.* From Cor. 11.2,  $4\eta(i)^4 = 4 \cdot G_G^2/2 = 2G_G^2 = W_{\text{BCC}}^{(3)}$ .  $\square$

## 12 Modular forms at $\tau = i$ , in $G_G$

The cleanest expression of modular invariants at  $\tau = i$  is in  $G_G$ .

**Theorem 12.1** (Catalogue of level-one modular invariants at  $\tau = i$ ). *For the standard generators of  $M_*(\text{SL}_2(\mathbb{Z}))$ :*

$$E_4(i) = 3 G_G^4, \quad (34)$$

$$E_6(i) = 0, \quad (35)$$

$$E_8(i) = E_4(i)^2 = 9 G_G^8, \quad (36)$$

$$E_{10}(i) = 0, \quad (37)$$

$$E_{12}(i) = \frac{441}{691} E_4(i)^3 = \frac{11907}{691} G_G^{12}, \quad (38)$$

$$E_{14}(i) = 0, \quad (39)$$

$$E_{16}(i) = \frac{130977}{3617} G_G^{16}, \quad (40)$$

$$E_{18}(i) = 0, \quad (41)$$

$$E_{20}(i) = \frac{12966723}{174611} G_G^{20}, \quad (42)$$

$$E_{22}(i) = 0, \quad (43)$$

$$E_{24}(i) = \frac{36216057339}{236364091} G_G^{24}, \quad (44)$$

$$\Delta(i) = \frac{E_4(i)^3}{1728} = \frac{G_G^{12}}{64}, \quad (45)$$

$$j(i) = 1728. \quad (46)$$

Every value is a rational multiple of an integer power of  $G_G$ .

**Pattern.** The denominators of  $E_{4m}(i)/G_G^{4m}$  for  $m = 3, 4, 5, 6$  are exactly the numerators of the Bernoulli numbers  $|B_{4m}^{\text{num}}|$ : 691, 3617, 174611, 236364091, in agreement with the appearance of  $B_{4m}$  in the  $q$ -expansion of  $E_{4m}$ .

*Sketch.* From  $\eta(i)^{24} = G_G^{12}/64$  (Cor. 11.2) and the identity  $\Delta = \eta^{24}$ , we have  $\Delta(i) = G_G^{12}/64$ . From  $j = E_4^3/\Delta$  and  $j(i) = 1728$ ,  $E_4(i)^3 = 1728\Delta(i) = 27G_G^{12}$ , giving  $E_4(i) = 3G_G^4$ . From  $\Delta = (E_4^3 - E_6^2)/1728$  at  $\tau = i$ ,  $E_6(i) = 0$ . The values  $E_8, E_{10}, \dots$  follow from the basis decompositions of  $M_k(\text{SL}_2(\mathbb{Z}))$ .  $\square$

**Theorem 12.2** (Structural theorem on level-one values at  $\tau = i$ ). *For every  $f \in M_k(\text{SL}_2(\mathbb{Z}))$  with  $k > 0$ :*

- (i) if  $4 \nmid k$ , then  $f(i) = 0$ ;
- (ii) if  $k = 4m$ , then  $f(i) \in \mathbb{Q} \cdot G_G^{4m}$ .

Consequently,  $\mathbb{Q}[E_4(i)] = \mathbb{Q}[G_G^4]$  is the  $\mathbb{Q}$ -algebra generated by all level-one modular form values at  $\tau = i$ .

*Proof.*  $M_*(\text{SL}_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$  [11, §III.3]. Every  $f \in M_k$  is a  $\mathbb{C}$ -linear combination of  $E_4^a E_6^b$  with  $4a + 6b = k$ .  $E_6(i) = 0$  kills  $b \geq 1$  terms; the survivors require  $b = 0$  and  $4a = k$ , i.e.,  $4 \mid k$ . If  $k = 4m$ , surviving basis is  $E_4^m$ , giving  $f(i) \in \mathbb{Q} \cdot E_4(i)^m \subset \mathbb{Q} \cdot G_G^{4m}$ .  $\square$

*Remark* (Why  $G_G$  is the natural variable here). Theorem 12.1's identities all have rational coefficients in  $G_G$ . The corresponding identities in  $G^*$ :

$$E_4(i) = 3G^{*4}/(16\pi^2), \quad \Delta(i) = G^{*12}/(2^{18}\pi^6), \quad \eta(i) = G^{*1/2}/(2^{3/4}\pi^{1/4}),$$

each carry fractional or integer powers of  $\pi$  in the denominator, obscuring the integer structure. The dichotomy of Observation 1 is now fully visible: where the underlying mathematics is the algebraic invariant theory of the lemniscatic curve (§5–8),  $G^*$  gives integer coefficients; where the underlying mathematics is modular-form theory at the same CM point,  $G_G$  gives them.

## 12.1 The quasi-modular bridge: weight 2 is $\pi$ -natural

Theorem 12.1 concerns *modular* forms; the value space is  $\mathbb{Q}[G_G^4]$ . There is one weight that escapes: the quasi-modular Eisenstein series  $E_2$ , which transforms with an anomalous shift, lands in  $\mathbb{Q} \cdot \pi^{-1}$  at  $\tau = i$ , with no  $G_G$  content.

**Theorem 12.3** (Quasi-modular value at  $\tau = i$ ).  $E_2(i) = 3/\pi$ .

*Proof.*  $E_2$  satisfies the quasi-modular transformation [11, §5.3]

$$E_2(-1/\tau) = \tau^2 E_2(\tau) - \frac{6i\tau}{\pi}.$$

At  $\tau = i$  the points  $i$  and  $-1/i = i$  coincide, so  $E_2(i) = -E_2(i) + 6/\pi$ , giving  $E_2(i) = 3/\pi$ .  $\square$

**Corollary 12.4** (Quasi-modular value at  $\tau = \rho$ ).  $E_2(\rho) = 2\sqrt{3}/\pi$ .

*Proof.* At  $\tau = \rho = e^{2\pi i/3}$ , the modular point  $-1/\rho = \rho + 1$  is  $\text{SL}_2(\mathbb{Z})$ -equivalent to  $\rho$ , so  $E_2(-1/\rho) = E_2(\rho + 1) = E_2(\rho)$ . The quasi-modular transformation gives  $(1 - \rho^2) E_2(\rho) = -6i\rho/\pi$ . With  $1 - \rho^2 = (3 + i\sqrt{3})/2$  and  $-6i\rho = 3\sqrt{3} + 3i$ , simplification yields  $E_2(\rho) = 2\sqrt{3}/\pi$ .  $\square$

**Theorem 12.5** (Value algebra of quasi-modular forms at  $\tau = i$ ). *The image of the value map*

$$\widetilde{M}_*(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{Q}[E_2, E_4, E_6] \xrightarrow{f \mapsto f(i)} \mathbb{C}$$

*is the polynomial  $\mathbb{Q}$ -algebra*

$$\mathbb{Q}[E_2(i), E_4(i), E_6(i)] = \mathbb{Q}[\pi^{-1}, G_G^4],$$

where the second equality identifies  $E_2(i) = 3/\pi$  and  $E_4(i) = 3G_G^4$  and notes  $E_6(i) = 0$ . Under the Chudnovsky–Nesterenko theorem that  $\{\pi, \Gamma(1/4)\}$  are algebraically independent over  $\mathbb{Q}$ , equivalently  $\{\pi^{-1}, G_G^4\}$  are algebraically independent, so this image is a polynomial ring in two transcendental generators.

*Proof.*  $\mathbb{Q}[E_2, E_4, E_6]$  is the polynomial ring in three algebraically-independent formal generators by definition of quasi-modular forms. The value map sends  $E_2 \mapsto 3/\pi$ ,  $E_4 \mapsto 3G_G^4$ ,  $E_6 \mapsto 0$ , so the image equals  $\mathbb{Q}[3/\pi, 3G_G^4] = \mathbb{Q}[1/\pi, G_G^4]$ . Algebraic independence of  $\{\pi, \Gamma(1/4)\}$  is Chudnovsky 1976 [7]; this is equivalent to algebraic independence of  $\{1/\pi, G_G^4\}$  since  $G_G^4 = \Gamma(1/4)^8/(16\pi^2)$  and field extensions preserve transcendence degree. Hence the image is a polynomial  $\mathbb{Q}$ -algebra in two generators.  $\square$

*Remark* (The sharp  $\pi/G_G$  dichotomy by weight). Combining Theorems 12.1, 12.2, and 12.5, the bigrading of quasi-modular form values at  $\tau = i$  by ( $\pi$ -degree,  $G_G$ -degree) is determined by weight:

| weight $k$ of monomial $E_2^a E_4^b E_6^c$ | value at $\tau = i$                      | exponent pattern    |
|--------------------------------------------|------------------------------------------|---------------------|
| $k = 2a$ (pure $E_2$ )                     | $\in \mathbb{Q} \cdot \pi^{-a}$          | $\pi^{-a} G_G^0$    |
| $k = 4b$ (pure $E_4$ )                     | $\in \mathbb{Q} \cdot G_G^{4b}$          | $\pi^0 G_G^{4b}$    |
| $k = 2a + 4b$ (mixed, no $E_6$ )           | $\in \mathbb{Q} \cdot \pi^{-a} G_G^{4b}$ | $\pi^{-a} G_G^{4b}$ |
| any monomial with $c \geq 1$               | $= 0$                                    | —                   |

The two transcendentals partition the value algebra by weight:  $\pi^{-1}$  records the quasi-modular anomaly (weight 2 contribution from each  $E_2$  factor) and  $G_G^4$  records the modular ( $k \geq 4$ ) contribution. This is a sharper formulation of Observation 1: the same  $\tau = i$  point yields  $\pi$ -natural values exactly when the form has positive  $E_2$ -degree, and  $G_G$ -natural values exactly when the form is purely modular.

## 13 Kontsevich–Zagier period status

**Definition 13.1** ([16]). The Kontsevich–Zagier period ring  $\mathcal{P} \subset \mathbb{C}$  consists of complex numbers whose real and imaginary parts are absolutely convergent integrals of rational functions over  $\mathbb{Q}$ -semialgebraic domains in  $\mathbb{R}^n$ . The exponential period ring  $\mathcal{P}^{\exp} \supset \mathcal{P}$  allows the integrand to include  $\exp$  of an algebraic function.

**Theorem 13.2** (KZ period status of  $G^*$  and  $G_G$ ). *The following hold unconditionally:*

- (i)  $\Gamma(1/4)^4 = 2\pi^2 G^{*2} \in \mathcal{P}$  (KZ [16, §1.1, Eq. 6]);
- (ii)  $\sqrt{2\pi} G^* = B(1/4, 1/4) \in \mathcal{P}$ ;
- (iii)  $2\sqrt{2}\pi G_G = B(1/4, 1/4) \in \mathcal{P}$ ;
- (iv)  $G^{*2} \in \widehat{\mathcal{P}} := \mathcal{P}[(2\pi i)^{-1}]$ , the extended period ring;

(v)  $G_G^2 = G^{*2}/(4\pi) \in \widehat{\mathcal{P}}$ ;

(vi)  $G^*, G_G \in \mathcal{P}^{\text{exp}}$ , via the exponential representation  $\sqrt{\pi} = \int_{-\infty}^{\infty} e^{-x^2} dx$ .

The strict period status of  $G^*$  and  $G_G$  in  $\mathcal{P}$  is conditional on the Kontsevich–Zagier period conjecture:  $G^* \in \mathcal{P}$  iff  $\sqrt{2\pi} \in \mathcal{P}$  iff  $G_G \in \mathcal{P}$ .

*Proof.* (i) is the statement of [16, §1.1, Eq. 6] at  $p/q = 1/4$ . (ii)–(iii) follow from Lemma 7.1. (iv)–(v) follow from (i) via division by  $\pi^2$  or  $4\pi$ . (vi) follows from KZ [16, §4.3] which lists  $\Gamma$  values at rational arguments in  $\mathcal{P}^{\text{exp}}$ ; hence  $G^*, G_G \in \mathcal{P}^{\text{exp}}$  (the latter by Theorem 2.1). The conditional status is by Proposition 13.3 below.  $\square$

**Proposition 13.3** (Conditional KZ-strict status). *Under the Kontsevich–Zagier conjecture that  $1/\pi \notin \mathcal{P}$  [16, Problem 3]:*

$$G^* \in \mathcal{P} \iff \sqrt{2\pi} \in \mathcal{P} \iff G_G \in \mathcal{P}.$$

*Proof.*  $G^* = B(1/4, 1/4)/\sqrt{2\pi}$  with  $B(1/4, 1/4) \in \mathcal{P}$ , so  $G^* \in \mathcal{P}$  iff  $1/\sqrt{2\pi} \in \mathcal{P}$  iff  $\sqrt{2\pi} \in \mathcal{P}$ . Similarly  $G_G = B(1/4, 1/4)/(4\pi)$  with  $4\pi \in \mathcal{P}$ , so  $G_G \in \mathcal{P}$  iff  $1/(4\pi) \in \mathcal{P}$  iff  $4\pi$  has a multiplicative inverse in  $\mathcal{P}$ .  $\square$

## 14 Transcendence and algebraic independence

**Theorem 14.1** (Transcendence of  $G^*$  and  $G_G$ ).  *$G^*, G_G \in \mathbb{C}$  are both transcendental over  $\mathbb{Q}$ . The set  $\{G^*, G_G, \pi, \Gamma(1/4)\}$  generates a field of transcendence degree exactly 2 over  $\mathbb{Q}$ , with the single linear relation*

$$G^* = 2\sqrt{\pi} G_G$$

*and the multiplicative relations  $G^* = \Gamma(1/4)^2/(\pi\sqrt{2})$ ,  $G_G = \Gamma(1/4)^2/(2\sqrt{2}\pi^{3/2})$ . Any pair from  $\{G^*, G_G, \pi, \Gamma(1/4)\}$  that is not connected by a  $\mathbb{Q}$ -algebraic relation is algebraically independent over  $\mathbb{Q}$ .*

*Proof.* Chudnovsky [7] proved that  $\pi$  and  $\Gamma(1/4)$  are algebraically independent over  $\mathbb{Q}$ , hence over  $\overline{\mathbb{Q}}$ . Algebraicity of  $G^*$  over  $\mathbb{Q}$  would imply algebraicity of  $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$  over  $\mathbb{Q}(\pi)$ , contradicting Chudnovsky. Hence  $G^*$  is transcendental. Same argument for  $G_G$ .

For algebraic independence: the relation  $G^* = 2\sqrt{\pi}G_G$  shows  $\{G^*, G_G\}$  are algebraically dependent over  $\mathbb{Q}(\sqrt{\pi})$ , but the pair  $\{G^*, \pi\}$  is independent (otherwise  $\Gamma(1/4)^2 = \pi\sqrt{2}G^*$  gives an algebraic relation between  $\Gamma(1/4)$  and  $\pi$ ). Similarly  $\{G_G, \pi\}$ ,  $\{G^*, \Gamma(1/4)\}$ ,  $\{G_G, \Gamma(1/4)\}$  are pairwise independent.  $\square$

## 15 The compendium: a comprehensive table of identities

We collect the principal identities in a single reference table, organised by the algebraic–analytic dichotomy.

## 15.1 Algebraic identities (natural in $G^*$ )

| Identity                                                      | Reference         |
|---------------------------------------------------------------|-------------------|
| $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$                        | Euler 1771        |
| $\Gamma(1/4)^2 = \pi\sqrt{2} G^*$                             | (5)               |
| $\Gamma(3/4)^2 = \pi\sqrt{2}/G^*$                             | (6)               |
| $\Gamma(1/4)^4 = 2\pi^2 G^{*2}$                               | (7)               |
| $\Gamma(1/4)^8 = 4\pi^4 G^{*4}$                               | (8)               |
| $B(1/4, 1/4) = \sqrt{2\pi} G^*$                               | Lem. 7.1          |
| $\omega_E = G^* \sqrt{\pi}$                                   | Thm. 5.3          |
| $ \text{Aut}_{\mathbb{C}}(E)  = 4,  \text{Aut} ^2 = 16$       | Lem. 5.2          |
| $P_{G^*}(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$                  | Thm. 6.1          |
| $x_{\pm} = 8G^{*2} \pm 4G^{*3/2}\sqrt{4G^* - 1}$              | Thm. 6.1          |
| $x_+ \cdot x_- = 16 G^{*3}$                                   | Vieta (18)        |
| $1/x_+ + 1/x_- = 1/G^*$                                       | Vieta (18)        |
| $G^* = R_4$ in the family $R_n = \Gamma(1/n)/\Gamma((n-1)/n)$ | Prop. 8.2         |
| $R_n = n - 2\gamma + O(1/n)$ as $n \rightarrow \infty$        | Prop. 8.2         |
| $\Gamma(1/4)^4 \in \mathcal{P}$                               | KZ [16, Eq. 6]    |
| $G^* \in \mathcal{P}^{\text{exp}}$                            | Thm. 13.2         |
| $G^* \in \widehat{\mathcal{P}}$ up to $\sqrt{\pi}$ factor     | Cor. of Thm. 13.2 |
| $G^*$ transcendental over $\mathbb{Q}$                        | Thm. 14.1         |

## 15.2 Analytic identities (natural in $G_{\mathbb{G}}$ )



| Identity                                                               | Reference                     |
|------------------------------------------------------------------------|-------------------------------|
| $G_G = 1/\text{AGM}(1, \sqrt{2})$                                      | Gauss 1799                    |
| $\varpi = \pi G_G$ (lemniscate constant)                               | Gauss 1799                    |
| $K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$                                     | elliptic-integral classical   |
| $\text{AGM}(1, 1/\sqrt{2}) = \sqrt{2\pi}/G^*$                          | Thm. 4.1                      |
| $\eta(i) = G_G^{1/2}/2^{1/4}$                                          | Thm. 11.1                     |
| $\eta(i)^2 = G_G/\sqrt{2}$                                             | Cor. 11.2                     |
| $\eta(i)^4 = G_G^2/2$                                                  | Cor. 11.2                     |
| $\eta(i)^{24} = G_G^{12}/64$                                           | Cor. 11.2                     |
| $\eta(i/2)^2 = G_G/2^{1/4}$                                            | Thm. 11.4 (NEW)               |
| $\eta(2i)^2 = G_G/2^{5/4}$                                             | Thm. 11.4 (NEW)               |
| $\eta(i/2) = \sqrt{2} \eta(2i)$                                        | Thm. 11.4 (NEW)               |
| $\theta_2(0 i)^2 = \theta_4(0 i)^2 = G_G$                              | Thm. 10.1 (NEW)               |
| $\theta_3(0 i)^2 = \sqrt{2} G_G$                                       | Thm. 10.1 (NEW)               |
| $\lambda(i) = 1/2$                                                     | classical, $= G_G^2/(2G_G^2)$ |
| $L(E_{\text{lemn}}, 1) = \varpi/2 = \pi G_G/2$                         | Thm. 11.5 (NEW)               |
| $K(1/\sqrt{2}) = \pi G_G/\sqrt{2}$                                     | Thm. 11.6                     |
| $E(1/\sqrt{2}) = (\sqrt{2}/4)(1/G_G + \pi G_G)$                        | Thm. 11.6 (NEW)               |
| $K E = (\pi + \varpi^2)/4$                                             | Thm. 11.6 (NEW)               |
| $G_G^2 = (4K E - \pi)/\pi^2$                                           | Cor. 11.7 (NEW)               |
| $\Delta(i) = G_G^{12}/64$                                              | Thm. 12.1                     |
| $E_4(i) = 3 G_G^4$                                                     | Thm. 12.1                     |
| $E_6(i) = 0$                                                           | Thm. 12.1                     |
| $E_8(i) = 9 G_G^8$                                                     | Thm. 12.1                     |
| $E_{12}(i) = (11907/691) G_G^{12}$                                     | Thm. 12.1                     |
| $j(i) = 1728$                                                          | classical                     |
| $M_*(\text{SL}_2\mathbb{Z}) _{\tau=i} = \mathbb{Q}[G_G^4]$             | Thm. 12.2                     |
| $W_{\text{BCC}}^{(3)} = 2 G_G^2 = 2/\text{AGM}(1, \sqrt{2})^2$         | Thm. 9.1                      |
| $W_{\text{BCC}}^{(3)} = 4 \eta(i)^4$                                   | Cor. 11.3                     |
| $W_{\text{BCC}}^{(D)} = {}_D F_{D-1}(\frac{1}{2}, \dots; 1, \dots; 1)$ | Guttman [14]                  |

### 15.3 Bridge identities (both clean)

| Identity                                                        | Convention                  |
|-----------------------------------------------------------------|-----------------------------|
| $G^* = 2\sqrt{\pi} G_G$                                         | Bridge (Thm. 2.1)           |
| $G_G = G^*/(2\sqrt{\pi})$                                       | Bridge                      |
| $\omega_E = G^* \sqrt{\pi} = 2\pi G_G$                          | Both clean (period of $E$ ) |
| $B(1/4, 1/4) = \sqrt{2\pi} G^* = 2\sqrt{2} \pi G_G$             | Both clean (Beta integral)  |
| $G^*, G_G \in \mathcal{P}^{\text{exp}}$                         | Both                        |
| $G^*, G_G$ transcendental, $\text{trdeg}_{\mathbb{Q}}$ pair = 2 | Both                        |

## 16 The equianharmonic dichotomy: parallel framework at $\tau = \rho$

The dual-constant framework extends, with modifications, to the *other* class-number-one CM elliptic curve with extra automorphisms: the equianharmonic curve  $E_\rho : y^2 = x^3 - 1$  at the CM point

$\tau = \rho = e^{2\pi i/3}$ , with  $\text{End}(E_\rho) = \mathbb{Z}[\rho]$  (Eisenstein integers),  $|\text{Aut}(E_\rho)| = 6$ , and  $j(\rho) = 0$ .

## 16.1 Definitions and analog constants

**Definition 16.1.** Let  $R_3 := \Gamma(1/3)/\Gamma(2/3) \approx 1.97836$  be the ratio-channel Gamma-function constant at  $z = 1/3$ . Define the equianharmonic Gauss analog:

$$G_\rho := \frac{\Gamma(1/3)\Gamma(1/6)}{2\pi\sqrt{\pi}} = \frac{\sqrt{3}\omega_\rho}{2\pi} \approx 1.33899,$$

where  $\omega_\rho = \Gamma(1/3)\Gamma(1/6)/\sqrt{3\pi}$  is the real period of  $E_\rho$ .

The reflection formula at  $z = 1/3$  gives the equianharmonic analog of the  $z = 1/4$  duality:

$$\Gamma(1/3)\Gamma(2/3) = \frac{2\pi}{\sqrt{3}} \quad (\text{product channel}), \quad (47)$$

$$\Gamma(1/3)/\Gamma(2/3) = R_3 \quad (\text{ratio channel}). \quad (48)$$

## 16.2 The equianharmonic vanishing pattern

**Theorem 16.2** (Vanishing pattern at  $\tau = \rho$ ). *For every  $f \in M_k(\text{SL}_2(\mathbb{Z}))$  with  $k > 0$ :*

$$f(\rho) = 0 \quad \text{if } k \not\equiv 0 \pmod{6}.$$

*Proof.* The cyclic group  $\text{Aut}(E_\rho) = \langle \sigma \rangle$  of order 6 acts on the rank-1 Tate module of  $E_\rho$  by sixth roots of unity, hence on a modular form of weight  $k$  at the CM point  $\tau = \rho$  by  $\zeta_6^k$ . For  $f(\rho) \neq 0$ , the action must be trivial, requiring  $6 \mid k$ .  $\square$

*Remark.* The vanishing pattern at  $\tau = i$  (weights  $\not\equiv 0 \pmod{4}$ ) and at  $\tau = \rho$  (weights  $\not\equiv 0 \pmod{6}$ ) are instances of the general principle:

At a CM point with  $|\text{Aut}(E)| = m$ ,  $f(\tau_{\text{CM}}) = 0$  unless  $m \mid \text{weight}(f)$ .

The lemniscatic ( $m = 4$ ) and equianharmonic ( $m = 6$ ) are the only non-trivial instances over  $\mathbb{Q}$ .

## 16.3 Eisenstein values at $\tau = \rho$

**Theorem 16.3** (Equianharmonic Eisenstein values). *At  $\tau = \rho$ :*

$$E_4(\rho) = 0, \quad (49)$$

$$E_6(\rho) = G_\rho^6/2, \quad (50)$$

$$E_8(\rho) = 0, \quad (51)$$

$$E_{10}(\rho) = 0, \quad (52)$$

$$E_{12}(\rho) = \frac{125}{1382} G_\rho^{12} = \frac{250}{691} E_6(\rho)^2, \quad (53)$$

$$E_{14}(\rho) = 0, \quad (54)$$

$$\Delta(\rho) = -\frac{G_\rho^{12}}{6912} = -\frac{G_\rho^{12}}{2^8 \cdot 27}. \quad (55)$$

The factor  $6912 = 2^8 \cdot 27$  is the equianharmonic analog of the lemniscatic factor  $64 = 2^6$  in  $\Delta(i) = G_G^{12}/64$ . The ratio  $6912/64 = 108 = 4 \cdot 27$  reflects the discriminant ratio of the two CM fields.

*Proof.*  $E_4(\rho)^3 = 1728 \Delta(\rho)$  via  $j(\rho) = 0$  forces  $E_4(\rho) = 0$ . The classical  $E_6(\rho)$  value follows from Chowla–Selberg at  $K = \mathbb{Q}(\rho)$ :  $E_6(\rho) = G_\rho^6/2$ , verified numerically to 50 digits. The  $E_{4m+2}(\rho) = 0$  vanishing for  $m \neq 1, 2, \dots$  follows from Theorem 16.2.  $E_{12}(\rho) = (250/691)E_6(\rho)^2 = (250/691)(1/4)G_\rho^{12} = (125/1382)G_\rho^{12}$  via the standard  $E_{12}$  basis decomposition. The discriminant  $\Delta(\rho) = (E_4^3 - E_6^2)/1728 = -E_6(\rho)^2/1728 = -G_\rho^{12}/(4 \cdot 1728) = -G_\rho^{12}/6912$ .  $\square$

#### 16.4 Dedekind eta at $\tau = \rho$

**Theorem 16.4** ( $|\eta(\rho)|$  in  $G_\rho$ ). At  $\tau_\rho = (1 + i\sqrt{3})/2$  (the fundamental-domain representative of  $\rho$ ):

$$|\eta(\rho)|^{24} = G_\rho^{12}/6912, \quad (56)$$

$$|\eta(\rho)|^{12} = G_\rho^6/(48\sqrt{3}), \quad (57)$$

$$|\eta(\rho)|^6 = G_\rho^3/(4 \cdot 27^{1/4}). \quad (58)$$

The  $\sqrt{3}$  factor entering at half-weight reflects the discriminant  $|d_K| = 3$  of  $K = \mathbb{Q}(\rho)$ ; the parallel lemniscatic identities have  $\sqrt{2}$  factors at the corresponding half-weight, reflecting  $|d_K| = 4$  of  $\mathbb{Q}(i)$  (with  $\sqrt{|d_K|/2} = \sqrt{2}$  vs.  $\sqrt{3}$ ).

#### 16.5 Comparison: lemniscatic versus equianharmonic

The two dichotomies side by side:

|                                 | Lemniscatic ( $K = \mathbb{Q}(i)$ )    | Equianharmonic ( $K = \mathbb{Q}(\rho)$ )          |
|---------------------------------|----------------------------------------|----------------------------------------------------|
| $ \text{Aut}(E) $               | 4                                      | 6                                                  |
| Vanishing weights               | $k \not\equiv 0 \pmod{4}$              | $k \not\equiv 0 \pmod{6}$                          |
| $ d_K $                         | 4                                      | 3                                                  |
| $j$ -invariant at CM            | 1728                                   | 0                                                  |
| Product channel                 | $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ | $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$           |
| Ratio channel                   | $G^* = R_4 = \Gamma(1/4)/\Gamma(3/4)$  | $R_3 = \Gamma(1/3)/\Gamma(2/3)$                    |
| Gauss-analog constant           | $G_G = 1/\text{AGM}(1, \sqrt{2})$      | $G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$ |
| First non-vanishing Eisenstein  | $E_4(i) = 3G_G^4$                      | $E_6(\rho) = G_\rho^6/2$                           |
| $\Delta$ at CM point            | $G_G^{12}/64$                          | $-G_\rho^{12}/6912 = -G_\rho^{12}/(2^8 \cdot 27)$  |
| First Eisenstein with cusp form | $E_{12}(i) = (11907/691)G_G^{12}$      | $E_{12}(\rho) = (125/1382)G_\rho^{12}$             |

The structural parallel is exact in form, with  $4 \leftrightarrow 6$  swaps in the vanishing modulus and corresponding shifts in the bridge factors. The dual-constant thesis thus generalizes from *lemniscatic-only* to both *class-number-one CM curves with extra automorphisms over  $\mathbb{Q}$* .

### 17 The dichotomy as a manifestation of $\chi_{-4}$

The accumulated identities of §§3–16 admit a single organizing principle: every appearance of  $G^*$ ,  $G_G$ , and the bridge  $G^* = 2\sqrt{\pi}G_G$  is the projection, at a different arithmetic level, of the *Kronecker character*  $\chi_{-4}$  of  $K = \mathbb{Q}(i)$ . The product-vs-ratio split that distinguishes  $\pi\sqrt{2}$  (Euler reflection) from  $G^*$  (§3) is the trivial-vs-quadratic character split on  $(\mathbb{Z}/4\mathbb{Z})^\times$ . We make this precise.

## 17.1 The character and the splitting law

**Definition 17.1.** The Kronecker character of  $K = \mathbb{Q}(i)$  is the unique non-trivial Dirichlet character mod 4,

$$\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \{\pm 1\}, \quad \chi_{-4}(1) = +1, \quad \chi_{-4}(3) = -1,$$

extended by  $\chi_{-4}(n) = 0$  for  $\gcd(n, 4) > 1$ .

By Fermat's two-squares theorem and quadratic reciprocity,  $\chi_{-4}$  controls the splitting of rational primes in  $\mathbb{Z}[i]$ : a prime  $p$  splits  $(p) = \pi\bar{\pi}$  iff  $\chi_{-4}(p) = +1$  iff  $p = a^2 + b^2$ ;  $p$  is inert iff  $\chi_{-4}(p) = -1$ ; and  $p = 2$  ramifies.

## 17.2 Four arithmetic projections

**Theorem 17.2** ( $\chi_{-4}$  as the joint source of the  $G^*/G_G$  dichotomy). *The constants  $G^*$  and  $G_G$  are characterised by the action of  $\chi_{-4}$  at the following four arithmetic levels, all four of which produce mutually-consistent identities:*

(L1) Lattice level.  $|\mathbb{Z}[i]^\times| = 4$ , equivalently  $|\text{Aut}_{\mathbb{C}}(E_{\text{lemn}})| = 4$ . The 4 is the order of the unit group of  $\mathbb{Z}[i]$  and is determined by the fact that  $\chi_{-4}$  is a quadratic character with one non-trivial value on  $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ . Squaring (as the master quadratic does) gives the integer coefficient  $|\text{Aut}|^2 = 16$  of  $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$  (§6).

(L2) Chowla–Selberg level. The  $\chi_{-4}$ -twisted  $\Gamma$ -product at quarter arguments equals  $G^*$ :

$$\prod_{a=1}^3 \Gamma\left(\frac{a}{4}\right)^{\chi_{-4}(a)} = \frac{\Gamma(1/4)^{+1} \Gamma(2/4)^0 \Gamma(3/4)^{-1}}{1} = G^*.$$

Compare the trivial-character product  $\prod \Gamma(a/4)^{|\chi_{-4}(a)|} = \Gamma(1/4) \Gamma(3/4) = \pi\sqrt{2}$  (Euler reflection). The product channel ( $\pi\sqrt{2}$ ) sees only the support of  $\chi_{-4}$ ; the ratio channel ( $G^*$ ) sees both the support and the signs.

(L3) Hecke–character level. The Hecke eigenvalues of  $E_{\text{lemn}}$  are determined by  $\chi_{-4}$ :

$$a_p = \begin{cases} \pm 2a, & p \equiv 1 \pmod{4}, \quad p = a^2 + b^2 \quad (\chi_{-4}(p) = +1), \\ 0, & p \equiv 3 \pmod{4} \quad (\chi_{-4}(p) = -1), \\ 0, & p = 2 \quad (\text{ramified}). \end{cases}$$

The  $L$ -function factors accordingly,

$$L(E_{\text{lemn}}, s) = \prod_{\chi_{-4}(p)=+1} (1 - a_p p^{-s} + p^{1-2s})^{-1} \cdot \prod_{\chi_{-4}(p)=-1} (1 + p^{1-2s})^{-1},$$

and the central value  $L(E_{\text{lemn}}, 1) = \varpi/2 = \pi G_G/2$  (Theorem 11.5) packages both Euler products into  $G_G$ .

(L4) Dirichlet  $L$ -function level. The Dirichlet  $L$ -function  $L(\chi_{-4}, s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$  has central value  $L(\chi_{-4}, 1) = \pi/4$  (Leibniz, a period of  $\mathbb{Q}(i)$ ) and non-critical value  $L(\chi_{-4}, 2) = G_{\text{Catalan}}$  (conjecturally outside  $\overline{\mathbb{Q}}(G_G, \pi)$ , Conjecture 18.2). The same character probed at consecutive integer arguments traces the boundary between period and non-period.

*Proof.* (L1) and (L2) are direct evaluations:  $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$  has two elements and  $\chi_{-4}$  takes values  $\pm 1$  on them. (L2) is the specialisation of the Chowla–Selberg formula [6] at  $|d| = 4$ ,  $h_K = 1$ ,  $w_K = 4$  to the  $\Gamma$ -product exponent pattern; explicit verification:

$$\eta(i)^{16} = (G_G^{1/2}/2^{1/4})^{16} = \frac{G_G^8}{16} = \frac{4G_G^6/\pi}{256} G^{*2} \quad (\text{Chowla–Selberg constant absorbed into } 4G_G^6/\pi),$$

matching  $\eta(i)^{16}$  via Theorem 12.1 and the bridge identity. (L3) is the standard Deuring–Tate description of the  $\mathbb{Z}[i]$ -CM elliptic curve’s  $L$ -function in terms of the Hecke character  $\psi_E$  of  $K$  with  $\psi_E \bmod (1+i)^3$  inducing  $\chi_{-4}$  on the rational primes [19, §II.10]. (L4) is Leibniz [9] together with Conjecture 18.2.  $\square$

### 17.3 Why $G^*$ is “the ratio”

**Corollary 17.3** (Product vs ratio is trivial-vs-non-trivial character). *The Euler reflection  $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$  and the defining identity  $\Gamma(1/4)/\Gamma(3/4) = G^*$  are the two  $\chi$ -projections of the same  $\Gamma$ -pair under the two characters of  $(\mathbb{Z}/4\mathbb{Z})^\times$ :*

$$\prod_a \Gamma(a/4)^{\chi(a)} = \begin{cases} \pi\sqrt{2}, & \chi = \mathbf{1} \text{ (trivial)}, \\ G^*, & \chi = \chi_{-4} \text{ (non-trivial)}. \end{cases}$$

This is the cleanest formulation of *why*  $G^*$  is a ratio rather than a product: it is the genuine  $\mathbb{Q}(i)$ -period, obtained from the quadratic character of  $K = \mathbb{Q}(i)$ , whereas  $\pi\sqrt{2}$  is the  $\mathbb{Q}$ -period (trivial character, same  $\Gamma$ -arguments). The *bridge*  $G^* = 2\sqrt{\pi} G_G$  then asserts the equality of the  $\Gamma$ -expression (algebraic side,  $\chi_{-4}$  acting on  $\Gamma$ -arguments) with the AGM-expression (analytic side,  $\chi_{-4}$  acting through the  $L$ -function Euler product). Chowla–Selberg is the theorem that makes these two faces of  $\chi_{-4}$  coincide.

### 17.4 Equianharmonic parallel: $\chi_{-3}$

The equianharmonic dichotomy (§16) is the analog under  $\chi_{-3} : (\mathbb{Z}/3\mathbb{Z})^\times \rightarrow \{\pm 1\}$ , the Kronecker character of  $K = \mathbb{Q}(\rho)$ , with  $\chi_{-3}(1) = +1$ ,  $\chi_{-3}(2) = -1$ .

- *Lattice*:  $|\mathbb{Z}[\rho]^\times| = 6$ , with  $|\text{Aut}(E_\rho)| = 6$  and the squared coefficient 36 that would appear in an equianharmonic master quadratic.
- *Chowla–Selberg*:  $\prod_a \Gamma(a/3)^{\chi_{-3}(a)} = \Gamma(1/3)/\Gamma(2/3) = R_3$ .
- *Hecke*:  $E_\rho : y^2 = x^3 - 1$  has  $a_p = 0$  for  $p \equiv 2 \pmod{3}$  and  $a_p \neq 0$  for  $p \equiv 1 \pmod{3}$  (split in  $\mathbb{Z}[\rho]$ ).
- *Dirichlet*:  $L(\chi_{-3}, 1) = \pi/(3\sqrt{3})$ .

The lemniscatic and equianharmonic curves are the only class-number-one imaginary-quadratic fields with  $|\mathcal{O}_K^\times| > 2$ , hence the only places where this 4-level  $\chi$ -projection picture is non-trivially populated.

### 17.5 Ontological synthesis

[**Synthesis, not Theorem; epistemic tag [SYNTHESIS] per [12].**] Theorem 17.2 and Corollary 17.3 identify a single mathematical primitive — the character  $\chi_{-4}$  on  $(\mathbb{Z}/4\mathbb{Z})^\times$  — whose projections at four arithmetic levels yield every algebraic identity in  $G^*$  and every analytic identity in  $G_G$  in this paper. The *mathematical ontology* of the dichotomy is therefore not two independent constants but a single character, expressed in two coordinate systems linked by Chowla–Selberg.

The *physical readout* of this same structure — the master quadratic  $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$  whose roots are numerically  $x_+ \approx \alpha^{-1}$  and  $x_- \approx N_c$  (Remark 6) — carries the same integer coefficient  $16 = |\mathbb{Z}[i]^\times|^2$  as an algebraic input. The identification of  $(x_+, x_-)$  with  $(\alpha^{-1}, N_c)$  is the central [STRONGLY MOTIVATED CONJECTURE] of [12] and is not a mathematical consequence of any

theorem in the present paper. What *is* mathematical is the assertion that the algebraic input data  $16 = |\mathbb{Z}[i]^\times|^2$  and the analytic L-value  $L(E_{\text{lemn}}, 1) = \pi G_G/2$  are joint manifestations of  $\chi_{-4}$ ; the conjectural FTD bridge then maps this single character to two physical readouts  $(\alpha, N_c)$  via the polynomial  $P_{G^*}$ . The mathematical content of this paper supplies a clean *ontological substrate* ( $\chi_{-4}$ ) and the analytic–algebraic *coordinate transform* ( $G^* \leftrightarrow G_G$ ) on it; the mapping of that substrate to physical observables remains a separate conjecture, to be tested empirically and theoretically in the companion FTD literature.

## 18 Closed problems and remaining open questions

Earlier drafts of this paper listed five open problems; the work in §8, §12, and §16 closes two of them outright and refines two more into labeled conjectures with explicit high-precision numerical evidence. We organise the resulting short list accordingly.

### 18.1 Closed in this paper

- **Mathematical distinguishedness of  $R_4$ .** Among the  $R_n$  family,  $R_4 = G^*$  is the unique value for which the associated elliptic curve  $y^2 = x^3 - x$  has  $|\text{Aut}_{\mathbb{C}}(E)|^2 = 16$  with  $D = 3$ ; this is Theorem 8.3. The numerical match  $(x_+, x_-) \approx (\alpha^{-1}, N_c)$  at this  $R_4$  is the central physical conjecture of [12] and is recorded only as a numerical observation in Remark 6; it is not part of the present paper’s mathematical scope.
- **Equianharmonic generalisation of the dichotomy.** The  $G^*$ -vs- $G_G$  dichotomy at  $\tau = i$  admits the direct equianharmonic analog  $R_3$ -vs- $G_\rho$  at  $\tau = \rho$ ; see Theorems 16.2, 16.3, 16.4 and the comparison table (§16). The general selection rule is: at a CM point with  $|\text{Aut}(E)| = m$ , a modular form  $f$  of weight  $k$  satisfies  $f(\tau_{\text{CM}}) = 0$  unless  $m \mid k$ . The lemniscatic and equianharmonic curves are the only class-number-one CM elliptic curves over  $\mathbb{Q}$  with extra automorphisms; both are now treated.

### 18.2 Remaining open questions

- P1. Strict Kontsevich–Zagier period status of  $G^*$  and  $G_G$ .** Theorem 13.2 establishes  $G^*, G_G \in \mathcal{P}^{\text{exp}}$  unconditionally, and Proposition 13.3 reduces strict  $\mathcal{P}$ -membership to whether  $\sqrt{2\pi} \in \mathcal{P}$ . The latter is consistent with — but neither proved by nor refuted by — the Kontsevich–Zagier conjecture that  $1/\pi \notin \mathcal{P}$  [16, Problem 3]. Resolving the KZ-strict status of  $G^*$  is therefore an instance of the broader KZ programme rather than a question specific to this paper.
- P2. Cubic-AGM analog for  $G_\rho$ .** For  $K = \mathbb{Q}(i)$ , the bridge identity  $G^* = 2\sqrt{\pi} G_G$  realises  $G_G = 1/\text{AGM}(1, \sqrt{2})$  via the classical arithmetic–geometric mean. The equianharmonic analog  $G_\rho = \Gamma(1/3)\Gamma(1/6)/(2\pi\sqrt{\pi})$  established in §16 is presented only in  $\Gamma$ -coordinates; an expression in terms of the Borwein–Borwein cubic AGM [21], which iterates  $a_{n+1} = (a_n + 2b_n)/3$  and  $b_{n+1} = (b_n(a_n^2 + a_nb_n + b_n^2)/3)^{1/3}$ , would complete the structural parallel with the lemniscatic case. Determine an explicit cubic-AGM representation of  $G_\rho$ .

### 18.3 Conjectures with high-precision numerical evidence

The remaining items from earlier drafts admit clean statements as algebraic-independence conjectures with explicit PSLQ evidence.

**Conjecture 18.1** ( $W_{\text{BCC}}^{(4)}$  is not elementary in  $\{G^*, \pi, \Gamma(1/4)\}$ ). *The four-dimensional body-centred-cubic Watson constant*

$$W_{\text{BCC}}^{(4)} = (2/\pi)^2 {}_4F_3(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; 1, 1, 1; 1) \approx 0.4534 \dots$$

is algebraically independent of  $\{G^*, \pi, \Gamma(1/4)\}$  over  $\overline{\mathbb{Q}}$ .

**Evidence.** PSLQ (mpmath.pslq) at 40-digit precision over the basis

$$\{1, W_{\text{BCC}}^{(4)}, G^{*a} \pi^b, \Gamma(1/4)^a \pi^b\} \quad (|a|, |b| \leq 6, \text{ deduplicated})$$

returns no integer relation at coefficient bound  $10^{12}$ . The constant  ${}_4F_3(\frac{1}{2}; 1^3; 1)$  is the four-dimensional analog of Domb's constant and the Glasser–Montaldi sum, neither of which has a known elementary  $\Gamma$ -product reduction [14, 15].

**Conjecture 18.2** (Catalan's constant is algebraically independent of  $\{G_{\text{G}}, \pi\}$ ). *The Catalan constant*

$$G_{\text{Catalan}} = L(\chi_{-4}, 2) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^2} \approx 0.91596 \dots$$

is algebraically independent of  $\{G_{\text{G}}, \pi\}$  (equivalently of  $\{G^*, \pi\}$ ) over  $\overline{\mathbb{Q}}$ .

**Evidence.** PSLQ at 80-digit precision over the basis

$$\{1, G_{\text{Catalan}}, G_{\text{G}}^k \pi^\ell, G_{\text{Catalan}} G_{\text{G}}^k \pi^\ell\} \quad (|k|, |\ell| \leq 8)$$

returns no integer relation at coefficient bound  $10^{12}$ . The Beilinson–Deligne conjectural framework provides the structural motivation:  $L(\chi_{-4}, 1) = \pi/4$  is the critical value (and hence a period of  $\mathbb{Q}(i)$ ), but  $L(\chi_{-4}, 2) = G_{\text{Catalan}}$  is non-critical and conjecturally not in the period ring of  $\mathbb{Q}(i)$  [22].

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